

One or Two Tanks?



Steve Greenberg is a facilities energy management engineer at Lawrence Berkeley National Laboratory and a technical editor at *Home Energy*.

Q. Is it cheaper for a homeowner to run one 100-gallon gas hot water tank or two 50-gallon tanks? There are six heavy water users in our household. How do you figure the cost? Do you include the outside surface heat loss and recovery rate? Do you just keep it simple by calculating what it takes to heat a gallon of cold water coming into the tank?

A Clean and Thirsty Household in Arizona

A. All other things being equal (recovery efficiency, height, and insulation level), one 100-gallon tank would use less gas than two 50-gallon tanks, because there's a lower surface-to-volume ratio with the single heater. But the open uninsulated flue of a standard water heater dominates the heat loss, and if that area is similar for the two options, then the standby loss would likewise be similar.

There are two variables of significance: the recovery efficiency and the standby loss. Both are included in the energy factor (EF), the official rating that comes from a test using a standard program of water use over a fixed period, with specified inlet and outlet temperatures. The recovery efficiency is the ratio of output to input energy and doesn't take into account off-cycle losses. First-hour rating is also very important; it indicates how much hot water a waiting heater can deliver (basically, an amount equal to the size of the tank plus the amount of water the unit can heat in an hour, which in turn is determined by the size of the burner and the recovery efficiency).

But if you're trying to minimize the cost of providing the amenity of hot water services, there may be other options to consider. First, reduce demand by using energy-saving showerheads and hot water fixtures and by washing clothes and dishes at the lowest temperature that is feasible. This should let you turn down the water heater thermostat, saving more energy by reducing standby loss. Second, meet the demand as efficiently as possible. Typical storage-type heaters have an EF of about 0.60–0.65, meaning that, using the standard test, they waste up to 40% of their input energy. The most efficient gas water heaters use condensing heat exchangers. These water heaters have recovery efficiencies of over 90% and an EF of up to 0.86. There's also the option of installing a tankless heater. These heaters have essentially no standby loss, but they must be sized for the largest instantaneous load, since their (large) burners must heat the water on demand. Tankless heaters typically have an EF in the low 0.80s.

To calculate the cost of domestic hot water, use the following equations:

Equation 1

$$(\text{Gallons of hot water}) \times (\text{Temperature difference } [^{\circ}\text{F}] \text{ between cold and hot water}) \times (8.34 \text{ Btu/gal } ^{\circ}\text{F}) / (100,000 \text{ Btu/therm}) = \text{Therms of load}$$

Equation 2

$$(\text{Therms of load}) / (\text{Overall efficiency}) = \text{Therms of gas}$$

Equation 3

$$(\text{Therms of gas}) \times (\text{Cost per therm}) = \text{Cost}$$

You can use EF for efficiency. But if you know the recovery efficiency and



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These antique water heaters are no match for many of today's large homes with heavy hot water use.

the standby loss, you can be more precise for your specific usage pattern: Use the above formula, substituting recovery efficiency for EF. Then add the standby loss to get the total usage.

You also might consider getting a solar hot water system, because you can meet most of your hot water needs using free solar energy, especially in climates like Arizona. Here's a pretty good go-to page: www1.eere.energy.gov/solar/sh_basics_water.html.

Also, you could look into a wastewater heat exchanger, especially if your main hot water load is from showers. Here is a helpful Web link: <http://gfxtechnology.com/>.



FOR MORE INFORMATION:

Some useful Web resources are Home Energy Saver (<http://hes.lbl.gov/hes>) and ACEEE (www.aceee.org/consumer-guide/topwater.htm).